

Aeroelastic properties of backward swept blades

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It is investigated how the nonlinear geometric coupling between torsion and flapwise bending of backward swept blades affects their static and dynamic aeroelastic properties. A geometrically nonlinear co-rotational finite beam element formulation coupled with the **Blade Element Momentum method** is used to compute the steady state blade deflection and aerodynamic power and thrust (assuming uniform wind and neglecting tower shadow and gravity effects) of a blade with different backward sweep shapes. A linearization of this aeroelastic model with the addition of an unsteady aerodynamic model is used to compute frequencies, damping and mode shapes of the aeroelastic blade modes from eigenvalue analysis, and to compute the transfer function from mean wind speed variations to variations in the blade root bending moments. Using the blade of the 5 MW NREL reference turbine as a baseline, it is shown that the backward sweep creates torsion towards feathering for downwind flapwise deflection in the first flapwise bending mode. This torsional component is shown to cause the frequency of the first aeroelastic flapwise bending mode to increase compared to the frequency of the corresponding structural mode, and this increase becomes larger for larger sweep. The frequency response of the flapwise blade root moment from wind excitation is shown to decrease below the increased first flapwise frequency which explains the reduced flapwise loads found in previous studies of backward swept blades. Computations of the classical flutter speed show that it decreases with the backward sweep.

I. Introduction

The structural and aeroelastic modal dynamics of backward swept blades is considered in this paper based on eigenvalue and frequency domain analyses of a linear aeroelastic blade model using co-rotational beam elements.

Backward sweep of wind turbine blades have been shown numerically and experimentally to reduce the fatigue and extreme blade root bending loads due to the load alleviating coupling between the flapwise bending and rotation of the chord towards feathering. Larwood and Zuteck¹ used the FAST code from NREL to show that a **28 m backward swept blade (STAR6) can have the similar, or only slightly higher flapwise blade root moments and maximum tip deflections as the original 25 m blade.** Due to the higher twist of the swept blade, the new rotor's 25 % larger area created only an estimated 5-10 % increase in energy capture, but with a blade with similar design loads. Ashwill *et al.*² have shown by field tests that a 26.1 m STAR prototype blade on a 54 m diameter rotor can increase the annual energy capture by 10-12 % compared to a baseline 23.5 m blade on a 48 m diameter Zond 750 kW turbine without increasing the blade root bending moments.

Riziotis *et al.*³ have found similar load alleviating results for backward swept blades using a nonlinear second order Bernoulli-Euler beam model and a multi-body code based on interconnected linear Bernoulli-Euler beam elements. They used the 5 MW NREL blade as baseline and showed a **5 % reduction in equivalent fatigue load for the flapwise blade root bending moment and a 8-12 % reduction in the mean value of the flapwise root bending moments** at around rated wind speeds. **The mean power was reduced about 2-4 %** at the same wind speeds compared to the original straight blade.

Verelst and Larsen⁴ investigated the effects of sweep on the fatigue and extreme loads and pitch travel for the 5 MW NREL blade. They have mapped the life-time equivalent fatigue values and extremes of blade root flapwise and edgewise bending and torsional loads in planes spanned by the sweep curve exponent and sweep offset at the tip. The flapwise load was decreased up to 10 % in fatigue and 15 % in extreme values, whereas the edgewise fatigue and extreme loads were increased by 6 %. The highest reductions in flapwise loads were for a sweep curve exponent of around 3. They also mapped the integrated pitch angle travel for both a Risø controller and the original NREL controller. The pitch travel was decreased up to 10 % for Risø controller by the sweep, whereas it was increased up to 8 % for NREL controller, but the pitch travel for the latter was still 10 % lower than for the Risø controller. They did not calculate or conclude on the reductions in annual energy production.

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This work focuses on understanding of the underlying static and dynamic effects that causes the passive load alleviation due to reduced angles of attack as a backward swept blade bends flapwise. The blade is modeled by linear Timoshenko beam elements in a nonlinear co-rotational formulation and coupled with the Blade Element Momentum (BEM) method for calculating the aerodynamic forces. The nonlinear steady state deflection of the blade is calculated for a given operational point assuming uniform inflow and zero gravity. The aeroelastic model including a model of unsteady aerodynamic forces is then linearized about the steady state to enable eigenvalue and frequency domain analyses of structural and coupled aeroelastic dynamics. Using the blade from the 5 MW NREL reference turbine as baseline, two backward swept blades are considered. The static bending–torsion coupling is shown to cause the steady state deflections to involve relative large rotations of the chord, whereby it will be necessary to pitch the swept blades at low wind speeds to track the optimal coefficient power. Computations of the structural and aeroelastic blade modes shows that the flapwise and edgewise blade bending modes contain torsion due to the inertia coupling from the backward sweep and large flapwise steady state deflection, respectively. The bending–torsion coupling due to the backward sweep increases the frequencies and decreases the aerodynamic damping of the flapwise blade modes. Computations of the transfer function from mean wind speed variations to blade root bending moments support the previous findings by showing decreased flapwise response and slightly increased edgewise response for frequencies below 3P where the excitation energy from rotational sampled turbulence can be assumed to be. Finally, it is shown that the relative tip speed where classical flutter occurs is reduced by the backward sweep.

II. Aeroelastic blade model

The aeroelastic model used in this investigation of blade aeroelasticity is the blade-part of a newly developed tool for aero-servo-elastic stability of the entire turbine called HAWCStab2, which is similar to the earlier version of called HAWCStab⁵, except that the kinematics are now based on a nonlinear co-rotational formulation⁶. This section contains a description of the aeroelastic modeling of a single blade in HAWCStab2.

A. Nonlinear co-rotational kinematics

The blade structure is modeled by discrete two-node prismatic Timoshenko elements that have their own coordinate systems with its origin in the first node (the innermost, closest to the rotor center) and the unit-vector of its third axis lying on the line from the first to the second node. The other axes of the element systems are rotated according to the principle axes of bending, with the first and second axes pointing towards the leading edge and towards the suction side/downwind, respectively. The degrees of freedom of the blade model are the three displacements and three rotations of the second node of each element, yielding a total of six times the number of elements N , typically $N = 20$.

The position vector $\mathbf{r}_{e,k}$ and orientation matrix $\mathbf{T}_{e,k}$ of element number k are written in the blade frame (azimuthal and blade pitch rotations are handled later) as

$$\begin{aligned}\mathbf{r}_{e,k} &= \sum_{n=1}^{k-1} \mathbf{T}_{e,n} (\delta \mathbf{u}_n + (l_n + \Delta l_n) \mathbf{e}_3) \\ \mathbf{T}_{e,k} &= \left(\prod_{n=1}^{k-1} \mathbf{E}_n \mathbf{T}[\delta \boldsymbol{\theta}_n] \right) \mathbf{E}_k\end{aligned}\quad (1)$$

where $\delta \mathbf{u}_n = \{\delta u_1 \ \delta u_2 \ \delta u_3\}_n^T$ and $\delta \boldsymbol{\theta}_n = \{\delta \theta_1 \ \delta \theta_2 \ \delta \theta_3\}_n^T$ are displacements and Euler angle rotations of the second node of element number n , which has the original (unloaded) length l_n and the elongation Δl_n due to the steady state deflection. The matrix $\mathbf{T}[\boldsymbol{\theta}] \equiv \mathbf{T}_1(\theta_1)\mathbf{T}_2(\theta_2)\mathbf{T}_3(\theta_3)$ defines the Euler angle rotations, where $\mathbf{T}_i$ describes rotation about the unit-vectors can be written in general form as

$$\mathbf{T}_i(\theta_i) = [\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3] \cos \theta_i + [\mathbf{e}_i \times \mathbf{e}_1 \ \mathbf{e}_i \times \mathbf{e}_2 \ \mathbf{e}_i \times \mathbf{e}_3] \sin \theta_i + (1 - \cos \theta_i) \mathbf{e}_i \mathbf{e}_i^T \quad (2)$$

where $\mathbf{e}_i$ are the unit-vectors $\mathbf{e}_1 = \{1 \ 0 \ 0\}^T$, $\mathbf{e}_2 = \{0 \ 1 \ 0\}^T$, and $\mathbf{e}_3 = \{0 \ 0 \ 1\}^T$.

The matrix $\mathbf{E}_k$ in Equation (1) is the element system of element number k given in the coordinates of the innermost neighboring element with number $k - 1$. The element system of the first element is placed in the rotor center $\mathbf{r}_{e,1} = \{0 \ 0 \ 0\}^T$ and its coordinate system rotated with the structural pitch (angle between the principle axis of flapwise bending and the rotor plane at zero blade pitch angle) around the unit-vector pointing from its first to second node.

Note that in a time simulation with the co-rotational formulation, the element systems $\mathbf{E}_k$ are continuously updated according to the displacements of the nodes. However, this update is only done in the present model during the

nonlinear iterative procedure to compute the steady state blade deflection as an equilibrium between internal blade forces and external aerodynamic forces. For linear eigenvalue analysis of small vibrations about the steady state deflection, the position of the elastic axis (the center of the principle axes of bending) and the rotation of a cross-section on element number k along the blade is written in the blade frame as

$$\begin{aligned}\mathbf{r}_{EA,k}(z; \mathbf{x}^s) &= \mathbf{r}_{e,k} + \mathbf{T}_{e,k} \left(\mathbf{u}_{0,k}(z) + \mathbf{u}_k(z; \mathbf{x}^s) + \{0 \ 0 \ z\}^T \right) \\ \mathbf{T}_{EA,k}(z; \mathbf{x}^s) &= \mathbf{T}_{e,k} \mathbf{T} [\boldsymbol{\theta}_{0,k}(z) + \boldsymbol{\theta}_k(z; \mathbf{x}^s)]\end{aligned}\quad (3)$$

where $z \in [0; l_k]$ is the cross-sectional coordinate along the element length. The displacements and rotations at the cross-section of the element are divided into steady state components and dynamic components given by

$$\begin{aligned}\begin{Bmatrix} \mathbf{u}_{0,k}(z) \\ \boldsymbol{\theta}_{0,k}(z) \end{Bmatrix} &= \mathbf{N}_k(z) \{ \boldsymbol{\theta}_{\text{node } 1,k}^T \ 0 \ 0 \ \Delta l_k \ \boldsymbol{\theta}_{\text{node } 2,k}^T \}^T \\ \begin{Bmatrix} \mathbf{u}_k(z; \mathbf{x}^s) \\ \boldsymbol{\theta}_k(z; \mathbf{x}^s) \end{Bmatrix} &= \mathbf{N}_k(z) \{ 0 \ 0 \ 0 \ \delta \mathbf{u}_k^T \ \delta \boldsymbol{\theta}_k^T \}^T\end{aligned}\quad (4)$$

where the matrix $\mathbf{N}_k(z)$ is the shape function matrix of the element, and the vectors $\boldsymbol{\theta}_{\text{node } 1,k}$ and $\boldsymbol{\theta}_{\text{node } 2,k}$ contain the steady state rotations at the first and second nodes of the element. Note that these element deformations are given in the element coordinate system with origin in the first node, whereby the shape function matrix $\mathbf{N}_k(z)$ becomes a 6×9 matrix. The shape functions used in this study are derived from static equilibrium of the prismatic two-node elements and include the effect of shear center misalignment with the elastic axis⁷. The structural degrees of freedom are collected in the vector $\mathbf{x}^s = \{ \delta \mathbf{u}_1^T \ \delta \boldsymbol{\theta}_1^T \ \dots \ \delta \mathbf{u}_N^T \ \delta \boldsymbol{\theta}_N^T \}^T$, which describes the dynamic displacements and rotations of the second node of each element away from the steady state.

Finally, the position of a point $\mathbf{r} = \{x, y, 0\}^T$ in the cross-sectional plane of element k can be written in the ground fixed frame as

$$\mathbf{r}_k(x, y, z; \mathbf{x}^s) = \mathbf{T}_\psi(t) \mathbf{E}_b (\mathbf{r}_{EA,k}(z; \mathbf{x}^s) + \mathbf{T}_{EA,k}(z; \mathbf{x}^s) \mathbf{r}(x, y)) \quad (5)$$

where $\mathbf{T}_\psi(t) = \mathbf{T}_2(\Omega t)$ handles the azimuthal rotation of the blade with the speed Ω about the global frame unit-vector $\mathbf{e}_2$ pointing down wind, $\mathbf{E}_b$ is the coordinate system of the blade frame rotated according to the blade pitch angle about the pitch axis which again may be rotated according to cone and delta-3 angles.

1. Relative inflow at a blade airfoil section

With the above definition of the nonlinear kinematics, it is possible to derive the relative inflow velocities at the aerodynamic calculation points along the blade span. Presently, there are aerodynamic calculations points at mid-point and second node of each element, except that there is no calculation point at the blade tip, where the aerodynamic forces are assumed to be zero (the same assumption is used for the rotor center point). The total number of calculation points are therefore $M = 2N - 1$.

Generally, the position of an aerodynamic calculation point number m is defined by a coordinate z_m on element number k_m . The inflow velocity vector in the ground fixed frame can therefore be derived as

$$\mathbf{V}_m(x, y; \mathbf{x}^s) = (V + v_a) \mathbf{e}_2 - v_t \mathbf{T}_\psi(t) \mathbf{e}_1 - \frac{d}{dt} (\mathbf{r}_{k_m}(x, y, z_m; \mathbf{x}^s)) \quad (6)$$

where V is the wind speed of the uniform inflow, and $v_{a,m}$ and $v_{t,m}$ are the steady state axial and tangential velocities induced by the wake at the aerodynamic calculation point m , which are assumed to be constant in this investigation, i.e., dynamic inflow effects are neglected by assuming frozen wake.

The relative inflow velocity vector in the local chord system can be derived as

$$\mathbf{v}_m(x, y; \mathbf{x}^s) = \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T(z_m; \mathbf{x}^s) \mathbf{E}_b^T \mathbf{T}_\psi^T \mathbf{V}_m \quad (7)$$

where $\mathbf{E}_{c,m}$ is the coordinate system of the local chord of the aerodynamic calculation point m given in the element system $\mathbf{E}_{k_m}$, which corresponds to rotation about the element length axis with the twist angle at the particular blade section minus the structural pitch of the element. Let $\mathbf{v}_m = \{v_{x,m}, v_{y,m}, v_{z,m}\}^T$, then note that $v_{x,m}$ is the chordwise velocity component (positive direction from leading to trailing edge), $v_{y,m}$ is the velocity component normal to the chord (positive towards the suction side), and $v_{z,m}$ is the spanwise cross-flow component.

Evaluation of the time derivative in (6) and substitution into (7) yields

$$\begin{aligned} \mathbf{v}_m = & (V - v_{a,m}) \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T \mathbf{E}_b^T \mathbf{e}_2 - v_{t,m} \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T \mathbf{E}_b^T \mathbf{e}_1 \\ & - \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T \mathbf{E}_b^T \boldsymbol{\omega} \times (\mathbf{E}_b (\mathbf{r}_{EA,k_m} + \mathbf{T}_{EA,k_m} \mathbf{r})) - \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T \left(\sum_{j=1}^{6N} \left(\frac{\partial}{\partial \dot{x}_j^s} (\mathbf{r}_{EA,k_m} + \mathbf{T}_{EA,k_m} \mathbf{r}) \right) \dot{x}_j^s \right) \end{aligned} \quad (8)$$

where x_j^s and $\dot{x}_j^s \equiv \partial x_j^s / \partial t$ are the structural degree of freedom number j and its time derivative, respectively, and it has been used that $\mathbf{T}_{\Psi}^T \mathbf{e}_2 = \mathbf{e}_2$, $\mathbf{T}_{\Psi}^T \mathbf{T}_{\Psi} = \mathbf{I}$, and $\mathbf{T}_{\Psi}^T \frac{\partial \mathbf{T}_{\Psi}}{\partial t} = \boldsymbol{\omega} \times ()$ with $\boldsymbol{\omega} = [0 \ \Omega \ 0]^T$.

The applied unsteady aerodynamic model⁸ requires the relative speed U_m and angle of attack α_m at the quarter chord point (the aerodynamic center) and the angle of attack $\alpha_{3/4,m}$ at the three-quarter chord point (the collocation point), which can be obtained by setting the particular values for x and y in the above equation. The vector $\mathbf{r}$ is therefore a vector from the elastic axis to either the aerodynamic center $\mathbf{r} = \mathbf{r}_{AC} = \{x_{AC}, y_{AC}, 0\}^T$, or collocation point at the three-quarter point $\mathbf{r} = \mathbf{r}_{3/4} = \{x_{3/4}, y_{3/4}, 0\}^T$ in the cross-section of the element system.

B. Linearized Lagrange equations

The equations of motion for the blade discretized by N Timoshenko elements can be derived from Lagrange's equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i^s} \right) - \frac{\partial L}{\partial x_i^s} + \frac{\partial D}{\partial \dot{x}_i^s} = Q_i \quad \text{for } i = 1, \dots, 6N \quad (9)$$

where $L = T - V$ is the Lagrangian given by the kinetic energy T and the potential energy V of the conservative forces, D is Rayleigh's dissipation function $D = \dot{\mathbf{x}}^s{}^T \mathbf{C} \dot{\mathbf{x}}^s$ given by the symmetric damping matrix $\mathbf{C}$ used to model the internal energy dissipation in the system as viscous damping, and Q_i is the generalized force for the generalized coordinate x_i^s due to the non-conservative aerodynamic forces, which in the present model are depending on the the blade displacements $\mathbf{x}^s$, velocities $\dot{\mathbf{x}}^s$, and $4M$ states of the aerodynamic model $\mathbf{x}^a$ (four states per aerodynamic calculation point). Note that the quadratic property of Rayleigh's dissipation function with a constant $\mathbf{C}$ matrix ensures linear damping forces in the equations of motion through the term $\partial D / \partial \dot{x}_i^s$ in Lagrange's equations.

The Lagrangian is a function of time, the generalized coordinates, and their time derivatives $L = L(t, \mathbf{x}^s, \dot{\mathbf{x}}^s)$. Using the chain rule, the first term of (9) can be expanded as

$$\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{x}_i^s} \right) + \sum_{k=1}^{6N} \frac{\partial^2 L}{\partial \dot{x}_i^s \partial \dot{x}_k^s} \ddot{x}_k^s + \sum_{k=1}^{6N} \frac{\partial^2 L}{\partial \dot{x}_i^s \partial x_k^s} \dot{x}_k^s - \frac{\partial L}{\partial x_i^s} + \frac{\partial D}{\partial \dot{x}_i^s} = Q_i \quad \text{for } i = 1, \dots, 6N \quad (10)$$

The steady state of the blade is described by the updated element coordinate systems $\mathbf{E}_k$ and the element deformations given by the elongation Δl_k and the node rotations $\boldsymbol{\theta}_{\text{node } 1,k}$ and $\boldsymbol{\theta}_{\text{node } 2,k}$, whereby the steady state is defined by $\mathbf{x}^s = \mathbf{0}$. Similar, the $4M$ aerodynamic states are defined such that $\mathbf{x}^a = \mathbf{0}$ in the steady state⁸. Taylor expansion of (10) for $\mathbf{x}^a = \mathbf{x}^s = \dot{\mathbf{x}}^s = \ddot{\mathbf{x}}^s = \mathbf{0}$ give the linear Lagrange's equations valid in the vicinity of the steady state:

$$\sum_{j=1}^{6N} \left((m_{ij} + m_{a,ij}) \ddot{x}_j^s + (g_{ij} + c_{ij} + c_{a,ij}) \dot{x}_j^s + (k_{ij} + k_{a,ij}) x_j^s \right) + \sum_{j=1}^{4M} k_{aa,ij} x_j^a = F_i \quad (11)$$

where c_{ij} are the elements of $\mathbf{C}$ and the coefficients due to inertia and conservative forces are

$$m_{ij} = \left. \frac{\partial^2 L}{\partial \dot{x}_i^s \partial \dot{x}_j^s} \right|_0, \quad g_{ij} = \left. \frac{\partial}{\partial t} m_{ij} + \left(\frac{\partial^2 L}{\partial \dot{x}_i^s \partial x_j^s} - \frac{\partial^2 L}{\partial x_i^s \partial \dot{x}_j^s} \right) \right|_0, \quad k_{ij} = \left. \frac{\partial}{\partial t} \left(\frac{\partial^2 L}{\partial \dot{x}_i^s \partial x_j^s} \right) - \frac{\partial^2 L}{\partial x_i^s \partial \dot{x}_j^s} \right|_0 \quad (12)$$

the coefficients due to the non-conservative aerodynamic forces are

$$m_{a,ij} = - \left. \frac{\partial Q_i}{\partial \ddot{x}_j^s} \right|_0, \quad c_{a,ij} = - \left. \frac{\partial Q_i}{\partial \dot{x}_j^s} \right|_0, \quad k_{a,ij} = - \left. \frac{\partial Q_i}{\partial x_j^s} \right|_0, \quad k_{aa,ij} = - \left. \frac{\partial Q_i}{\partial x_j^a} \right|_0 \quad (13)$$

and the steady state part of conservative and non-conservative forces F_i are

$$F_i = F_i^{\text{int}}|_0 + Q_i|_0 \quad (14)$$

where the internal force of the conservative forces is

$$F_i^{\text{int}}|_0 = \frac{\partial L}{\partial x_i^s} \Big|_0 - \frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{x}_i^s} \Big|_0 \right) \quad (15)$$

where the subscript “0” denotes the evaluation of the terms at the steady state $\mathbf{x}^a = \mathbf{x}^s = \dot{\mathbf{x}}^s = \mathbf{0}$. Note that the steady state is later computed by solving the equation $F_i = 0$ iteratively.

1. System matrices and internal forces from conservative forces

The total kinetic energy is given by an integral of the kinetic energy of each particle over the entire volume of the blade, which using the position vector in the ground fixed frame (5) can be written as a sum over all elements

$$T = \sum_{k=1}^N \int_0^{l_k} \int_{A_k} \frac{1}{2} \rho \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k dA_k dz \quad (16)$$

where A_k is the cross-sectional area of element k , and ρ is the material density which here is constant over the assumed isotropic cross-section of each prismatic element. The velocity vector $\dot{\mathbf{r}}_k$ can be expanded to

$$\dot{\mathbf{r}}_k = \frac{d\mathbf{r}_k}{dt} = \sum_{n=1}^N \frac{\partial \mathbf{r}_k}{\partial x_n^s} \dot{x}_n + \frac{\partial \mathbf{r}_k}{\partial t} \quad (17)$$

from which the following properties of the position and velocity vectors can be shown

$$\frac{\partial \dot{\mathbf{r}}_k}{\partial \dot{x}_i^s} = \frac{\partial \mathbf{r}_k}{\partial x_i^s} \quad \text{and} \quad \frac{\partial^2 \dot{\mathbf{r}}_k}{\partial \dot{x}_i^s \partial \dot{x}_j^s} = \frac{\partial \mathbf{r}_k}{\partial x_i^s \partial x_j^s} = \frac{\partial^2 \mathbf{r}_k}{\partial x_i^s \partial x_j^s} \quad (18)$$

Substituting the Lagrangian $L = T - V$ with (16) into the coefficients (12) and using the properties (18), it is seen that the coefficients of the linear equations of motion can be computed as

so simple

$$\left\{ \begin{aligned} m_{ij} &= \sum_{k=1}^N \int_0^{l_k} \int_{A_k} \rho \left(\frac{\partial \mathbf{r}_k^T}{\partial x_i^s} \frac{\partial \mathbf{r}_k}{\partial x_j^s} \right) \Big|_0 dA_k dz \\ g_{ij} &= \frac{\partial}{\partial t} m_{ij} + \sum_{k=1}^N \int_0^{l_k} \int_{A_k} \rho \left(\frac{\partial \mathbf{r}_k^T}{\partial x_i^s} \frac{\partial}{\partial x_j^s} \left(\frac{\partial \mathbf{r}_k}{\partial t} \right) - \frac{\partial \mathbf{r}_k^T}{\partial x_j^s} \frac{\partial}{\partial x_i^s} \left(\frac{\partial \mathbf{r}_k}{\partial t} \right) \right) \Big|_0 dA_k dz \\ k_{ij} &= \sum_{k=1}^N \int_0^{l_k} \int_{A_k} \rho \left(\frac{\partial \mathbf{r}_k^{2T}}{\partial x_i^s \partial x_j^s} \frac{\partial^2 \mathbf{r}_k}{\partial t^2} + \frac{\partial \mathbf{r}_k^T}{\partial x_i^s} \frac{\partial}{\partial x_j^s} \left(\frac{\partial^2 \mathbf{r}_k}{\partial t^2} \right) \right) \Big|_0 dA_k dz + \frac{\partial^2 V}{\partial x_i^s \partial x_j^s} \Big|_0 \end{aligned} \right. \quad (19)$$

where it is noted that only the partial time derivative of the position vector is needed, not the full velocity vector. Equation (19) shows that the mass matrix with elements m_{ij} is symmetric, and that the potential energy part of the stiffness matrix with elements s_{ij} is also symmetric. Because the position vector (5) has an explicit time dependency due to the azimuthal rotation of the blade, the gyroscopic matrix with elements g_{ij} has a symmetric part given by $\partial m_{ij} / \partial t$ and an anti-symmetric part. The stiffness matrix has a symmetric and an asymmetric part, where the symmetric part is the linear generalized force due to the centripetal acceleration of the blade, which due to the co-rotational formulation will correspond to the centrifugal stiffening term.

Substituting the Lagrangian $L = T - V$ with (16) into (15), the internal forces can be derived as

$$F_i^{\text{int}}|_0 = \sum_{k=1}^N \int_0^{l_k} \int_{A_k} \rho \left(\frac{\partial \mathbf{r}_k^T}{\partial x_i^s} \frac{\partial^2 \mathbf{r}_k}{\partial t^2} \right) \Big|_0 dA_k dz + \frac{\partial V}{\partial x_i^s} \Big|_0 \quad (20)$$

where the first term are the centrifugal forces and the second term are the internal elastic forces.

The evaluations of the integrals over the isotropic element cross-sections, the following relationships are used

$$\begin{aligned} \int_{A_k} \rho dA_k &= M, & \int_{A_k} \rho x dA_k &= M x_{CG}, & \int_{A_k} \rho y dA_k &= M y_{CG} \\ \int_{A_k} \rho x^2 dA_k &= M r_{y,EA}^2, & \int_{A_k} \rho y^2 dA_k &= M r_{x,EA}^2, & \int_{A_k} \rho xy dA_k &= M I_{xy,EA} \end{aligned} \quad (21)$$

where for each element k the parameter M is the mass per unit-length, x_{CG} and y_{CG} are the center of mass in the element system, $r_{x,EA}$ and $r_{y,EA}$ are the radii of gyration about the x and y axes of the element system (with the elastic axis as origin), and $I_{xy,EA}$ describes any inertia coupling between flapwise and edgewise bending, which is zero for a truly isotropic cross-section; however, the distribution of stiffness and mass over the cross-section may be defined as different with this term.

The total potential energy from elastic restoring forces in the Timoshenko element can be written as

$$V = \sum_{k=1}^N \int_0^{l_k} V_k(z) dz \quad (22)$$

where the potential elastic energy from a cross-section

$$\begin{aligned} V_k(z) = & \frac{1}{2} \left(\left(\frac{\partial \mathbf{u}_{0,k}^T(z)}{\partial z} + \frac{\partial \mathbf{u}_k^T(z; \mathbf{x}^s)}{\partial z} \right) \begin{bmatrix} k_x GA & 0 & 0 \\ 0 & k_y GA & 0 \\ 0 & 0 & EA \end{bmatrix} \left(\frac{\partial \mathbf{u}_{0,k}(z)}{\partial z} + \frac{\partial \mathbf{u}_k(z; \mathbf{x}^s)}{\partial z} \right) \right. \\ & + \left(\frac{\partial \boldsymbol{\theta}_{0,k}^T(z)}{\partial z} + \frac{\partial \boldsymbol{\theta}_k^T(z; \mathbf{x}^s)}{\partial z} \right) \begin{bmatrix} EI_x & 0 & 0 \\ 0 & EI_y & 0 \\ 0 & 0 & GK \end{bmatrix} \left(\frac{\partial \boldsymbol{\theta}_{0,k}(z)}{\partial z} + \frac{\partial \boldsymbol{\theta}_k(z; \mathbf{x}^s)}{\partial z} \right) \\ & + (\boldsymbol{\theta}_{0,k}^T(z) + \boldsymbol{\theta}_k^T(z; \mathbf{x}^s)) \begin{bmatrix} 0 & 0 & 0 \\ 0 & k_x GA & 0 \\ 0 & 0 & k_y GA \end{bmatrix} (\boldsymbol{\theta}_{0,k}(z) + \boldsymbol{\theta}_k(z; \mathbf{x}^s)) \\ & + \left(\frac{\partial \mathbf{u}_{0,k}^T(z)}{\partial z} + \frac{\partial \mathbf{u}_k^T(z; \mathbf{x}^s)}{\partial z} \right) \begin{bmatrix} 0 & -k_x GA & 0 \\ k_y GA & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} (\boldsymbol{\theta}_{0,k}(z) + \boldsymbol{\theta}_k(z; \mathbf{x}^s)) \\ & \left. + (\boldsymbol{\theta}_{0,k}^T(z) + \boldsymbol{\theta}_k^T(z; \mathbf{x}^s)) \begin{bmatrix} 0 & k_y GA & 0 \\ -k_x GA & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \left(\frac{\partial \mathbf{u}_{0,k}(z)}{\partial z} + \frac{\partial \mathbf{u}_k(z; \mathbf{x}^s)}{\partial z} \right) \right) \end{aligned} \quad (23)$$

where the steady state and dynamic displacements and rotations at the cross-section are given by (4), E and G are Young's modulus and the shear modulus, A is the cross-sectional area, I_x and I_y are area moments of inertia for flapwise and edgewise bending, K is the polar moment of inertia, and k_x and k_y are the shear factors of element k . Note that the co-rotational formulation (4) causes the blade degrees of freedom to be uncoupled in elastic stiffness.

2. System matrices and external forces from non-conservative forces

The generalized external force on coordinate x_i^s due to the aerodynamic forces on the blade is written as

$$Q_i = \sum_{k=1}^N \int_0^{l_k} \oint_{S_k(z)} \frac{\partial \mathbf{r}_k^T(x, y, z; \mathbf{x}^s)}{\partial x_i^s} \mathbf{P}_k(x, y, z) dS_k(z) dz \quad (24)$$

where $\mathbf{P}_k(x, y, z) dS_k(z)$ is the aerodynamic force vector normal to the airfoil surface at the element coordinate z on element k of the blade, and the inner closed curve integral runs along the entire surface of the airfoil section.

The force vector is defined in the ground fixed frame and can be expressed by the pressure force distribution $\mathbf{p}_k(x, y, z)$ in the chord coordinate system at z on element k as

$$\mathbf{P}_k(x, y, z) = \mathbf{T}_\psi \mathbf{E}_b \mathbf{T}_{EA,k}(z; \mathbf{x}^s) \mathbf{E}_{c,k}(z) \mathbf{p}_k(x, y, z) \quad (25)$$

where $\mathbf{E}_{c,k}(z)$ is the element coordinate systems along element k given the element system $\mathbf{E}_k$. Insertion of (25) and the position vector (5) into Equation (24), the generalized force becomes

$$Q_i = \sum_{k=1}^N \int_0^{l_k} \oint_{S_k(z)} \left(\frac{\partial \mathbf{r}_{EA,k}^T(z; \mathbf{x}^s)}{\partial x_i^s} + (\mathbf{r}_c^T \mathbf{E}_{c,k}^T(z) + \mathbf{r}_{AC}^T(z)) \frac{\partial \mathbf{T}_{EA,k}^T(z; \mathbf{x}^s)}{\partial x_i^s} \right) \mathbf{T}_{EA,k}(z; \mathbf{x}^s) \mathbf{E}_{c,k}(z) \mathbf{p}_k(x, y, z) dS_k(z) dz \quad (26)$$

where $\mathbf{r}_{AC}$ is the vector from the elastic axis to the aerodynamic center (AC), and $\mathbf{r}_c^T = \{x \ y \ 0\}$ is a vector from the AC to the airfoil surface given in the chord system, whereby the closed curve integration over x and y along the airfoil surface follows the standard definition for calculating the aerodynamic normal and tangential forces and the moment:

$$\mathbf{f}_k(z) = \oint_{S_k(z)} \mathbf{p}_k(x, y, z) dS_k(z) \quad \text{and} \quad m_k(z) \mathbf{e}_3 = \oint_{S_k(z)} \mathbf{r}_c \times \mathbf{p}_k(x, y, z) dS_k(z) \quad (27)$$

A part of the inner closed curve integral of (26) can thereby be evaluated as

$$Q_i = \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\frac{\partial \mathbf{r}_{EA,k}^T(z; \mathbf{x}^s)}{\partial x_i^s} + \mathbf{r}_{AC}^T(z) \frac{\partial \mathbf{T}_{EA,k}^T(z; \mathbf{x}^s)}{\partial x_i^s} \right) \mathbf{T}_{EA,k}(z; \mathbf{x}^s) \mathbf{E}_{c,k}(z) \mathbf{f}_k(z) + \oint_{S_k(z)} \mathbf{r}_c^T \mathbf{E}_{c,k}^T(z) \frac{\partial \mathbf{T}_{EA,k}^T(z; \mathbf{x}^s)}{\partial x_i^s} \mathbf{T}_{EA,k}(z; \mathbf{x}^s) \mathbf{E}_{c,k}(z) \mathbf{p}_k(x, y, z) dS_k(z) \right\} dz \quad (28)$$

To evaluate the second part of the inner closed curve integral it is necessary to rearrange the integrant. Let the rotational transformation matrix $\mathbf{T}_{EA,k}(z; \mathbf{x}^s)$ from the blade frame to the cross-section of the element k given by (1) and (3) be rewritten in the short form

$$\mathbf{T}_{EA,k}(z; \mathbf{x}^s) = \left(\prod_{n=1}^{k-1} \mathbf{E}_n \mathbf{T}[\theta_n] \right) \mathbf{E}_k \mathbf{T}[\theta_{0,k}(z) + \theta_k(z; \mathbf{x}^s)] = \prod_{n=1}^{6k} \mathbf{T}[x_n] \quad (29)$$

where the transformation matrix $\mathbf{T}[x_n]$ equals the identity matrix for translational degrees of freedom and otherwise also includes the necessary rotations according to the element orientations. Using this short form, the integrant of the second part of the inner closed curve integral of (28) can be rewritten as

$$\begin{aligned} \mathbf{r}_c^T \mathbf{E}_{c,k}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial x_i^s} \mathbf{T}_{EA,k} \mathbf{E}_{c,k} \mathbf{p}_k &= \mathbf{r}_c^T \mathbf{E}_{c,k}^T \left(\prod_{n=6k}^{i+1} \mathbf{T}^T[x_n] \right) \frac{\partial \mathbf{T}^T[x_i^s]}{\partial x_i^s} \left(\prod_{n=i-1}^1 \mathbf{T}^T[x_n] \right) \left(\prod_{n=1}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{p}_k \\ &= \left(\frac{\partial \mathbf{T}[x_i^s]}{\partial x_i^s} \left(\prod_{n=i+1}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{r}_c \right) \cdot \left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{p}_k \right) \\ &= \left(\boldsymbol{\omega}_i \times \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{r}_c \right) \cdot \left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{p}_k \right) \end{aligned} \quad (30)$$

where $\boldsymbol{\omega}_i$ is the unit-vector about which the rotation described by the degree of freedom x_i^s is defined. Using the equality $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ and additional rules from vector calculus, the above can be rearranged as

$$\begin{aligned} \mathbf{r}_c^T \mathbf{E}_{c,k}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial x_i^s} \mathbf{T}_{EA,k} \mathbf{E}_{c,k} \mathbf{p}_k &= \boldsymbol{\omega}_i \cdot \left(\left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{r}_c \right) \times \left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{p}_k \right) \right) \\ &= \boldsymbol{\omega}_i \cdot \det \left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \right) \left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \right)^{-T} \mathbf{r}_c \times \mathbf{p}_k \\ &= \boldsymbol{\omega}_i \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \mathbf{r}_c \times \mathbf{p}_k \end{aligned} \quad (31)$$

where the equalities $\det \left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \right) = 1$ and $\left(\left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k} \right)^{-T} = \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{E}_{c,k}$ have been used. Using this expression and the definition (27), the generalized force from the aerodynamic forces (28) can be written as

$$Q_i = \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial x_i^s} + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial x_i^s} \right) \mathbf{T}_{EA,k} \mathbf{E}_{c,k} \mathbf{f}_k(z) + \boldsymbol{\omega}_i \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \mathbf{e}_{3,c,k} m_k(z) \right\} dz \quad (32)$$

where $\mathbf{e}_{3,c,k}$ is the third unit-vector of the chord coordinate system described in the element system of element k .

The sectional aerodynamic forces $\mathbf{f}_k(z) = \mathbf{f}_k(z; \mathbf{x}^s, \dot{\mathbf{x}}^s, \mathbf{x}^a)$ and moment $m_k(z) = m_k(z; \mathbf{x}^s, \dot{\mathbf{x}}^s, \mathbf{x}^a)$ are functions of the blade displacements $\mathbf{x}^s$, velocities $\dot{\mathbf{x}}^s$ and the aerodynamic state $\mathbf{x}^a$ of the unsteady aerodynamic model⁸, which does

not include added mass effects from accelerations. The elements of the system matrices (13) due to aerodynamic forces can therefore be generally written as

$$\begin{aligned}
m_{a,ij} &= 0 \\
c_{a,ij} &= \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial \dot{x}_i^s} \right) \Big|_0 + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial \dot{x}_i^s} \Big|_0 \right) \mathbf{T}_{EA,k} \Big|_0 \mathbf{E}_{c,k} \frac{\partial \mathbf{f}_k}{\partial \dot{x}_j^s} \Big|_0 + \boldsymbol{\omega}_i \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} \frac{\partial m_k}{\partial \dot{x}_j^s} \Big|_0 \right\} dz \\
k_{a,ij} &= \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\left(\frac{\partial \mathbf{r}_{EA,k}^{2T}}{\partial \dot{x}_i^s \partial \dot{x}_j^s} \right) \Big|_0 + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^{2T}}{\partial \dot{x}_i^s \partial \dot{x}_j^s} \Big|_0 \right) \mathbf{T}_{EA,k} \Big|_0 + \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial \dot{x}_i^s} \right) \Big|_0 + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial \dot{x}_i^s} \Big|_0 \right) \frac{\partial \mathbf{T}_{EA,k}}{\partial \dot{x}_j^s} \Big|_0 \mathbf{E}_{c,k} \mathbf{f}_k \Big|_0 \right. \\
&\quad + \left(\frac{\partial \boldsymbol{\omega}_i}{\partial \dot{x}_j^s} \right) \Big|_0 \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} + \boldsymbol{\omega}_i \Big|_0 \cdot \frac{\partial}{\partial \dot{x}_j^s} \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} \Big) m_k \Big|_0 \\
&\quad \left. + \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial \dot{x}_i^s} \right) \Big|_0 + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial \dot{x}_i^s} \Big|_0 \right) \mathbf{T}_{EA,k} \Big|_0 \frac{\partial \mathbf{f}_k}{\partial \dot{x}_j^s} \Big|_0 + \boldsymbol{\omega}_i \Big|_0 \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} \frac{\partial m_k}{\partial \dot{x}_j^s} \Big|_0 \right\} dz \quad (33) \\
k_{aa,ij} &= \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial \dot{x}_i^s} \right) \Big|_0 + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial \dot{x}_i^s} \Big|_0 \right) \mathbf{T}_{EA,k} \Big|_0 \frac{\partial \mathbf{f}_k}{\partial \dot{x}_j^a} \Big|_0 + \boldsymbol{\omega}_i \Big|_0 \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} \frac{\partial m_k}{\partial \dot{x}_j^a} \Big|_0 \right\} dz
\end{aligned}$$

and steady state external forces $Q_i|_0$ become

$$Q_i|_0 = \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial \dot{x}_i^s} \right) \Big|_0 + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial \dot{x}_i^s} \Big|_0 \right) \mathbf{T}_{EA,k} \Big|_0 \mathbf{f}_k \Big|_0 + \boldsymbol{\omega}_i \Big|_0 \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} m_k \Big|_0 \right\} dz \quad (34)$$

All integrals are evaluated over the entire blade using cubic spline interpolation of the steady state forces (obtained from BEM solutions) for the computation of the generalized steady state forces (34), and linear interpolation of the derivatives of the aerodynamic forces and moment between aerodynamic calculation points for the computation of the system matrices (33). The derivatives of the aerodynamic forces at an aerodynamic calculation point m with respect to the structural displacements and velocities and the states of the aerodynamic model are derived in the following section.

C. Linearized unsteady aerodynamic model

The unsteady aerodynamic model⁸ contains four states for each aerodynamic calculation point; two states to describe the effect of shed vorticity using Jones' approximation to Wagner's function⁹, and two states to describe the time lags in the trailing edge separation point movement as in the Leishman-Beddoes dynamic stall model. Linearization of this model for small variation of the four aerodynamic states about the steady state defined by $\mathbf{x}_m^a$ yields⁸

$$\dot{\mathbf{x}}_m^a + \mathbf{A}_{d,m} \mathbf{x}_m^a = \mathbf{T}_{x\alpha_{3/4},m} \delta\alpha_{3/4} + \mathbf{T}_{xU,m} \delta U + \mathbf{T}_{x\dot{\alpha},m} \delta\dot{\alpha} \quad (35)$$

where $\delta\alpha_{3/4}$ denotes small variation of the angle of attack at three-quarter chord point from its steady state value α_0 , $\delta\dot{\alpha}$ denotes a small pitch rate, and δU denotes small variations of the relative inflow speed away from its steady state U_0 . The matrices and vectors of (35) for each aerodynamic calculation point are (omitting the subscript " m ")

$$\begin{aligned}
\mathbf{A}_d &= \begin{bmatrix} T_1^{-1} & 0 & 0 & 0 \\ 0 & T_2^{-1} & 0 & 0 \\ -T_p^{-1} C_{L,\alpha} & -T_p^{-1} C_{L,\alpha} & T_p^{-1} & 0 \\ 0 & 0 & -T_f^{-1} \frac{df^{st}}{d\alpha} \Big|_0 C_{L,\alpha}^{-1} & T_f^{-1} \end{bmatrix} \\
\mathbf{T}_{x\alpha_{3/4}} &= \begin{bmatrix} T_1^{-1} A_1 \\ T_2^{-1} A_2 \\ (1 - A_1 - A_2) T_p^{-1} C_{L,\alpha} \\ 0 \end{bmatrix}, \quad \mathbf{T}_{xU} = \begin{bmatrix} -\frac{A_1 \alpha_0}{U_0} \\ -\frac{A_2 \alpha_0}{U_0} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{T}_{x\dot{\alpha}} = \begin{bmatrix} 0 \\ 0 \\ \pi T_p^{-1} T_0 \\ 0 \end{bmatrix} \quad (36)
\end{aligned}$$

where $A_1 = 0.165$ and $A_2 = 0.335$ are amplitudes of Jones' approximation, $T_0 = c/(2U_0)$ is the half-chord travel time constant, $T_i = T_0/b_i$ with $b_1 = 0.0455$ and $b_2 = 0.3$ are time constants of Jones' approximation, T_p is the time

constant for lag in the pressure distribution (here set to $1.7T_0$), and T_f is the time constant for lag in the boundary layer separation (here set to $6.0T_0$). The constant $C_{L,\alpha}$ is the linear slope for attached flow ($\approx 2\pi$), and $df^{\text{st}}/d\alpha|_0$ is the slope of the static separation point function evaluated at the steady state angle of attack (see further details in reference⁸).

The small variations of inflow angles and relative speed can be expressed in terms of variations of inflow velocities as

$$\delta\alpha = \mathbf{T}_{\alpha v} \delta \mathbf{v}, \quad \delta\alpha_{3/4} = \mathbf{T}_{\alpha v} \delta \mathbf{v}_{3/4}, \quad \text{and} \quad \delta U = U_0^{-1} [v_{0x} \ v_{0y} \ 0] \delta \mathbf{v} \quad (37)$$

with

$$\mathbf{T}_{\alpha v} = U_0^{-2} [v_{0y} \ -v_{0x} \ 0] \quad (38)$$

where v_{0x} and v_{0y} are the steady state inflow velocities at the aerodynamic calculation point, and $\delta \mathbf{v}$ and $\delta \mathbf{v}_{3/4}$ are the variations of the inflow vector at the aerodynamic center and the three-quarter point, respectively.

Aerodynamic forces and moment in the chord coordinate system at aerodynamic calculation point m are given by

$$\begin{Bmatrix} \mathbf{f} \\ m \end{Bmatrix}_m = \frac{1}{2} \rho c_m U_m^2 \begin{bmatrix} \sin \alpha_m & -\cos \alpha_m & 0 \\ \cos \alpha_m & \sin \alpha_m & 0 \\ 0 & 0 & c_m \end{bmatrix} \begin{Bmatrix} C_L^{\text{dyn}} \\ C_D^{\text{dyn}} \\ C_M^{\text{dyn}} \end{Bmatrix}_m = \frac{1}{2} \rho c_m U_m \begin{bmatrix} v_{y,m} & v_{x,m} & 0 \\ -v_{x,m} & v_{y,m} & 0 \\ 0 & 0 & c_m U_m \end{bmatrix} \begin{Bmatrix} C_L^{\text{dyn}} \\ C_D^{\text{dyn}} \\ C_M^{\text{dyn}} \end{Bmatrix}_m \quad (39)$$

where c_m is the chord length, α_m is the angle of attack, U_m is the relative speed, and $\mathbf{v}_m = \{v_{x,m} \ v_{y,m} \ v_{z,m}\}^T$ is the relative inflow vector (7) at the calculation point. The unsteady aerodynamic coefficients can be approximated by a linear expression for small variations of the four aerodynamic states $\delta \mathbf{x}^a$, the inflow angles $\delta\alpha$ and $\delta\alpha_{3/4}$, and the pitch rate $\delta\dot{\alpha}$ about their steady state values (in general form omitting the subscript “ m ”):

$$\begin{Bmatrix} C_L^{\text{dyn}} \\ C_D^{\text{dyn}} \\ C_M^{\text{dyn}} \end{Bmatrix} \approx \begin{Bmatrix} C_{L,0} \\ C_{D,0} \\ C_{M,0} \end{Bmatrix} + \mathbf{T}_{C\mathbf{x}} \mathbf{x}^a + \mathbf{T}_{C\alpha_{3/4}} \delta\alpha_{3/4} + \mathbf{T}_{C\alpha} \delta\alpha + \mathbf{T}_{C\dot{\alpha}} \delta\dot{\alpha} \quad (40)$$

where $C_{L,0} = C_L^{\text{st}}(\alpha_0)$, $C_{D,0} = C_D^{\text{st}}(\alpha_0)$, and $C_{M,0} = C_M^{\text{st}}(\alpha_0)$ are the steady state lift, drag, and moment coefficients, and

$$\mathbf{T}_{C\mathbf{x}} = \begin{bmatrix} c_{l,\alpha} & c_{l,\alpha} & 0 & c_{l,f} \\ c_{d,\alpha} - C_{L,0} & c_{d,\alpha} - C_{L,0} & 0 & c_{d,f} \\ c_{m,\alpha} & c_{m,\alpha} & 0 & 0 \end{bmatrix} \quad \mathbf{T}_{C\alpha_{3/4}} = \begin{Bmatrix} c_{l,\alpha}(1 - A_1 - A_2) \\ (c_{d,\alpha} - C_{L,0})(1 - A_1 - A_2) \\ c_{m,\alpha}(1 - A_1 - A_2) \end{Bmatrix}, \quad \mathbf{T}_{C\alpha} = \begin{Bmatrix} 0 \\ C_{L,0} \\ 0 \end{Bmatrix}, \quad \mathbf{T}_{C\dot{\alpha}} = \begin{Bmatrix} \pi T_0 \\ 0 \\ -\frac{\pi}{2} T_0 \end{Bmatrix} \quad (41)$$

where the coefficients can be found in the reference⁸. Insertion of (40) with (37) into (39) and Taylor expansion for small variations of inflow velocities yields the linear approximation to the chord forces and moment :

$$\begin{Bmatrix} \mathbf{f} \\ m \end{Bmatrix} \approx \begin{Bmatrix} \mathbf{f}_0 \\ m_0 \end{Bmatrix} + \frac{1}{2} \rho c (\mathbf{T}_{f\mathbf{x}} \mathbf{x}^a + \mathbf{T}_{fv} \delta \mathbf{v} + \mathbf{T}_{fv_{3/4}} \delta \mathbf{v}_{3/4} + \mathbf{T}_{f\dot{\alpha}} \delta \dot{\alpha}) \quad (42)$$

where the matrices can be written as

$$\mathbf{T}_{f\mathbf{x}} = \mathbf{T}_{\text{tmp},1} \mathbf{T}_{C\mathbf{x}}, \quad \mathbf{T}_{fv} = \mathbf{T}_{\text{tmp},1} \mathbf{T}_{C\alpha} \mathbf{T}_{\alpha v} + \mathbf{T}_{\text{tmp},2}, \quad \mathbf{T}_{fv_{3/4}} = \mathbf{T}_{\text{tmp},1} \mathbf{T}_{C\alpha_{3/4}} \mathbf{T}_{\alpha v}, \quad \mathbf{T}_{f\dot{\alpha}} = \mathbf{T}_{\text{tmp},1} \mathbf{T}_{C\dot{\alpha}} \quad (43)$$

where the helping matrices $\mathbf{T}_{\text{tmp},1}$ and $\mathbf{T}_{\text{tmp},2}$ are

$$\mathbf{T}_{\text{tmp},1} = U_0 \begin{bmatrix} v_{0y} & v_{0x} & 0 \\ -v_{0x} & v_{0y} & 0 \\ 0 & 0 & cU_0 \end{bmatrix} \quad \mathbf{T}_{\text{tmp},2} = \begin{bmatrix} (v_{0y} C_{L,0} + v_{0x} C_{D,0}) v_{0x} / U_0 + U_0 C_{D,0} & (v_{0y} C_{L,0} + v_{0x} C_{D,0}) v_{0y} / U_0 + U_0 C_{L,0} & 0 \\ (v_{0y} C_{D,0} - v_{0x} C_{L,0}) v_{0x} / U_0 - U_0 C_{L,0} & (v_{0y} C_{D,0} - v_{0x} C_{L,0}) v_{0y} / U_0 + U_0 C_{D,0} & 0 \\ 2c v_{0x} C_{M,0} & 2c v_{0y} C_{M,0} & 0 \end{bmatrix} \quad (44)$$

The small variations of inflow velocities can be derived from (8) in general form as

$$\delta \mathbf{v}_m = \sum_{n=1}^{6N} \frac{\partial \mathbf{v}_m}{\partial x_j^s} \bigg|_0 x_j^s + \sum_{n=1}^{6N} \frac{\partial \mathbf{v}_m}{\partial \dot{x}_j^s} \bigg|_0 \dot{x}_j^s \quad (45)$$

where the velocity derivatives are

$$\begin{aligned} \frac{\partial \mathbf{v}_m}{\partial x_j^s} \bigg|_0 x_j^s &= (V - v_{a,m}) \mathbf{E}_{c,m}^T \frac{\partial \mathbf{T}_{EA,k_m}^T}{\partial x_j^s} \bigg|_0 \mathbf{E}_b^T \mathbf{e}_2 - v_{t,m} \mathbf{E}_{c,m}^T \frac{\partial \mathbf{T}_{EA,k_m}^T}{\partial x_j^s} \bigg|_0 \mathbf{E}_b^T \mathbf{e}_1 \\ &\quad - \mathbf{E}_{c,m}^T \frac{\partial \mathbf{T}_{EA,k_m}^T}{\partial x_j^s} \bigg|_0 \mathbf{E}_b^T \boldsymbol{\omega} \times (\mathbf{E}_b (\mathbf{r}_{EA,k_m}|_0 + \mathbf{T}_{EA,k_m}|_0 \mathbf{r})) \\ &\quad - \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T|_0 \mathbf{E}_b^T \boldsymbol{\omega} \times \left(\mathbf{E}_b \left(\frac{\partial \mathbf{r}_{EA,k_m}}{\partial x_j^s} \bigg|_0 + \frac{\partial \mathbf{T}_{EA,k_m}}{\partial x_j^s} \bigg|_0 \mathbf{r} \right) \right) \end{aligned} \quad (46)$$

$$\frac{\partial \mathbf{v}_m}{\partial \dot{x}_j^s} \bigg|_0 = -\mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T|_0 \left(\frac{\partial \mathbf{r}_{EA,k_m}}{\partial x_j^s} \bigg|_0 + \frac{\partial \mathbf{T}_{EA,k_m}}{\partial x_j^s} \bigg|_0 \mathbf{r} \right) \quad (47)$$

where $\mathbf{r} = \mathbf{r}_{AC} = \{x_{AC}, y_{AC}, 0\}^T$, or $\mathbf{r} = \mathbf{r}_{3/4} = \{x_{3/4}, y_{3/4}, 0\}^T$ depending on the cross-sectional point in which the velocity derivative is required. The variation of the pitch rate at the aerodynamic calculation point m can be derived as

$$\delta \dot{\alpha} = \sum_{n=1}^{6N} \left(\mathbf{e}_1 \cdot \mathbf{E}_{c,k}^T \mathbf{T}_{EA,k}^T|_0 \frac{\partial \mathbf{T}_{EA,k}}{\partial x_j^s} \bigg|_0 \mathbf{E}_{c,k} \mathbf{e}_2 \right) \dot{x}_j^s \quad (48)$$

Insertion of (45) and (48) into (42) yields the derivatives of the aerodynamic forces and moment can be derived as:

$$\begin{aligned} \frac{\partial}{\partial x_j^s} \left\{ \frac{\mathbf{f}}{m} \right\}_m &= \frac{1}{2} \rho c_m \left(\mathbf{T}_{fv} \frac{\partial \mathbf{v}_m}{\partial x_j^s} \bigg|_0 + \mathbf{T}_{fv_{3/4}} \frac{\partial \mathbf{v}_{3/4,m}}{\partial x_j^s} \bigg|_0 \right) \\ \frac{\partial}{\partial \dot{x}_j^s} \left\{ \frac{\mathbf{f}}{m} \right\}_m &= \frac{1}{2} \rho c_m \left(\mathbf{T}_{fv} \frac{\partial \mathbf{v}_m}{\partial \dot{x}_j^s} \bigg|_0 + \mathbf{T}_{fv_{3/4}} \frac{\partial \mathbf{v}_{3/4,m}}{\partial \dot{x}_j^s} \bigg|_0 + \mathbf{T}_{f\dot{\alpha}} \left(\mathbf{e}_1 \cdot \mathbf{E}_{c,k}^T \mathbf{T}_{EA,k}^T|_0 \frac{\partial \mathbf{T}_{EA,k}}{\partial x_j^s} \bigg|_0 \mathbf{E}_{c,k} \mathbf{e}_2 \right) \right) \\ \frac{\partial}{\partial x_j^a} \left\{ \frac{\mathbf{f}}{m} \right\}_m &= \frac{1}{2} \rho c_m \mathbf{T}_{fx} \frac{\partial}{\partial x_j^a} \mathbf{x}_m^a \end{aligned} \quad (49)$$

where the velocity derivatives are computed using (46). Note that these derivatives are computed in all aerodynamic calculation points to obtain a distribution along the blade for each state variable (displacements, velocities and aerodynamic state variables), whereby the integrals of the coefficients (33) of the aerodynamic system matrices can be computed using linear interpolation between the calculation points.

III. Methods

A. Nonlinear steady state computation

The nonlinear steady state given by the orientation matrices element coordinate systems $\mathbf{E}_k$ and the element deformations (elongation Δl_k and node rotations $\boldsymbol{\theta}_{\text{node } 1,k}$ and $\boldsymbol{\theta}_{\text{node } 2,k}$) is obtained for each operational point given by the wind speed, rotor speed and pitch angle in an iterative process. First, the steady forces and moments in all aerodynamic calculation points are computed by BEM for the undeflected blade. Then, the blade is deflected until there is an equilibrium of these external aerodynamic forces and moments with the internal elastic forces due to the element deformations ($F_i = 0$, cf. Equation (14)). Then, the aerodynamic forces are updated by a new BEM computation followed by an update of the blade deflection. This iterative process continues until the blade tip movement in an update of its deflection is less than 0.01 % of the blade length.

B. Eigenvalue analysis

For each operation point, the structural coefficients (19) with (22) and aerodynamic coefficients (33) with (49) are computed using Gauss integration of analytical functions of the position vectors and rotation matrices and their derivatives. The linear equations of structural motion (11) and the aerodynamic state equations (35) with (37) are thereafter

collected in coupled aeroelastic equations

$$\begin{aligned} \mathbf{M}\ddot{\mathbf{x}}^s + (\mathbf{C} + \mathbf{G} + \mathbf{C}_a)\dot{\mathbf{x}}^s + (\mathbf{K} + \mathbf{K}_a)\mathbf{x}^s + \mathbf{K}_f\mathbf{x}^a &= \mathbf{0} \\ \dot{\mathbf{x}}^a + \mathbf{A}_d\mathbf{x}^a + \mathbf{C}_{sa}\dot{\mathbf{x}}^s + \mathbf{K}_{sa}\mathbf{x}^s &= \mathbf{0} \end{aligned} \quad (50)$$

which describes the aeroelastic blade response for small vibrations about the steady state equilibrium. Introduction of a state variable vector

$$\mathbf{y} = \begin{Bmatrix} \mathbf{x}^s \\ \dot{\mathbf{x}}^s \\ \mathbf{x}^a \end{Bmatrix} \quad (51)$$

and the homogenous solution $\mathbf{y} = \mathbf{w}e^{\lambda t}$, the following ordinary eigenvalue problem is set up

$$\left(\begin{bmatrix} \mathbf{0} & -\mathbf{I} & \mathbf{0} \\ \mathbf{M}^{-1}(\mathbf{K} + \mathbf{K}_a) & \mathbf{M}^{-1}(\mathbf{C} + \mathbf{G} + \mathbf{C}_a) & \mathbf{M}^{-1}\mathbf{A}_f \\ \mathbf{K}_{sa} & \mathbf{C}_{sa} & \mathbf{A}_d \end{bmatrix} + \lambda \mathbf{I} \right) \mathbf{w} = \mathbf{0} \quad (52)$$

An eigenvalue λ_n and corresponding eigenvector $\mathbf{w}_n$ define the aeroelastic modal frequencies, damping, and mode shapes of the blade. The natural frequency and logarithmic decrement for mode number n are given by the imaginary and real part of the eigenvalues

$$f_n = \frac{\text{Im}\{\lambda_n\}}{2\pi} \quad \text{and} \quad \delta_n = -\frac{\text{Re}\{\lambda_n\}}{f_k} \quad (53)$$

where the damping analogy to logarithmic decrement here assumes that the damping is small. This assumption is not fulfilled for highly damped modes (especially flapwise bending modes), however, the logarithmic decrement is applied for traditional reasons and should just be considered as the damping ratio multiplied by 2π .

C. Transfer functions from mean wind speed to blade root bending moments

As an example of the possibilities of frequency domain analysis, the excitation of the structural and aerodynamic state variables due to a small mean speed variation $\delta V(t)$ is added to the right-hand-side of (50) as

$$\begin{aligned} \mathbf{M}\ddot{\mathbf{x}}^s + (\mathbf{C} + \mathbf{G} + \mathbf{C}_a)\dot{\mathbf{x}}^s + (\mathbf{K} + \mathbf{K}_a)\mathbf{x}^s + \mathbf{K}_f\mathbf{x}^a &= \mathbf{B}_V^s \delta V(t) \\ \dot{\mathbf{x}}^a + \mathbf{A}_d\mathbf{x}^a + \mathbf{C}_{sa}\dot{\mathbf{x}}^s + \mathbf{K}_{sa}\mathbf{x}^s &= \mathbf{B}_V^a \delta V(t) \end{aligned} \quad (54)$$

where the elements of the column matrices $\mathbf{B}_V^s$ and $\mathbf{B}_V^a$ using (8), (33), and (35) can be written as

$$\begin{aligned} b_{V,i}^s &= \sum_{k=1}^N \int_0^{l_k} \left\{ \left(\frac{\partial \mathbf{r}_{EA,k}^T}{\partial x_i^s} \right) + \mathbf{r}_{AC}^T \frac{\partial \mathbf{T}_{EA,k}^T}{\partial x_i^s} \right\} \mathbf{T}_{EA,k}|_0 \frac{\partial \mathbf{f}_k}{\partial V} \Big|_0 + \boldsymbol{\omega}_i|_0 \cdot \left(\prod_{n=i}^{6k} \mathbf{T}[x_n] \right) \Big|_0 \mathbf{e}_{3,c,k} \frac{\partial m_k}{\partial V} \Big|_0 dz \\ \mathbf{b}_{V,m}^a &= (\mathbf{T}_{x\alpha_{3/4}} \mathbf{T}_{\alpha v} + \mathbf{T}_{xU} [v_{0x}/U_0 \quad v_{0y}/U_0 \quad 0]) \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T|_0 \mathbf{E}_b^T \mathbf{e}_2 \end{aligned} \quad (55)$$

where $\mathbf{b}_{V,m}^a$ are the four elements corresponding to the four aerodynamic state variables of calculation point m . The integrals of $b_{V,i}^s$ are evaluated using linear interpolation between calculation points, where the wind speed derivatives of the aerodynamic forces and moment at each aerodynamic calculation point m are computed as

$$\frac{\partial}{\partial V} \left\{ \mathbf{f} \right\}_m = \frac{1}{2} \rho c_m (\mathbf{T}_{fv} + \mathbf{T}_{fv_{3/4}}) \mathbf{E}_{c,m}^T \mathbf{T}_{EA,k_m}^T|_0 \mathbf{E}_b^T \mathbf{e}_2 \quad (56)$$

The frequency response function of the aeroelastic blade state variables due to a harmonic unit mean wind variation is

$$\begin{Bmatrix} \mathbf{X}^s(\omega) \\ \dot{\mathbf{X}}^s(\omega) \\ \mathbf{X}^a(\omega) \end{Bmatrix} = \left(\begin{bmatrix} \mathbf{0} & -\mathbf{I} & \mathbf{0} \\ \mathbf{M}^{-1}(\mathbf{K} + \mathbf{K}_a) & \mathbf{M}^{-1}(\mathbf{C} + \mathbf{G} + \mathbf{C}_a) & \mathbf{M}^{-1}\mathbf{A}_f \\ \mathbf{K}_{sa} & \mathbf{C}_{sa} & \mathbf{A}_d \end{bmatrix} + i\omega \mathbf{I} \right)^{-1} \begin{Bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{B}_V^s \\ \mathbf{B}_V^a \end{Bmatrix} \quad (57)$$

where ω is the frequency of the harmonic wind speed variations. The frequency response for the blade root bending moments is then obtained by multiplying the part of the elastic stiffness matrix belonging to the element with its first node at the blade root flange with the six components of $\mathbf{X}^s(\omega)$ describing the frequency response at the second node of this element (note that all elements are uncoupled in elastic stiffness forces due to the co-rotational formulation).

IV. Results

The baseline of this investigation is the 61.5 m blade of the 5 MW NREL reference turbine¹⁰, except that the centers of gravity (CG) along its span have been moved to the elastic axis to isolate the effects of the backward sweep. **This forward movement of CG will increase the flutter speed relative to the original blade¹¹.** The blade is mounted on a stiff hub with an outer radius of 1.5 m.

Two different blade sweeps are considered which are described by both linear and quadratic shape functions. The latter gives the backward sweep curvature leading to the beneficial structural coupling between flapwise bending and torsion. A linear forward sweep is added to compensate for an otherwise large steady state pitch moment:

$$\mathbf{r}_{EA} = \left\{ a \frac{z}{R_0} - b \left(\frac{z}{R_0} \right)^2, 0, z \right\}^T \quad z \in [0; R_0] \quad (58)$$

where z is the coordinate along the pitch axis of the blade, $R_0 = 61.5$ m is the outer radius, and a and b are the coefficients of the linear and quadratic terms. The sweep angle relative to the pitch axis is

$$\Lambda = \arctan \left(\frac{a}{R_0} - \frac{2bz}{R_0^2} \right) \quad (59)$$

The blade shapes and sweep angles are plotted in Figure 1, where the parameters for swept blade 1 are $a = 4$ and $b = 6$, and for swept blade 2 are $a = 8$ and $b = 12$. The nodes of the 20 Timoshenko beam elements are also plotted to show the distribution and to illustrate the rotation in the rotor plane of the chord coordinate systems of the aerodynamic calculation points, which are placed in the middle and at the second node of each element, except at the tip where the aerodynamic forces and moment are set to zero. The structural properties of the prismatic elements and the aerodynamic twist, chord and profile coefficients in the aerodynamic calculation points are the same of all three blades. A spectral damping model¹² is used to set the structural damping for the first two standstill blade modes to 1.8 % and 3.2 % logarithmic decrements for all three blades, and the damping of subsequent modes are increased 10 % relative to the previous mode with a maximum modeled damping ratio of 70 %.

A. Steady state operation

Figure 2 shows selected results of the nonlinear computation of the steady state, where the optimal tip speed ratio is presumed to be 8.7 with 6 rpm and 12.1 rpm as the lower and upper rotor speed limits. The plotted optimal pitch angle curves are obtained for each blade design by numerical optimization using the full nonlinear steady state computation scheme. The largest differences in pitch angles occur where the blade loadings are largest. The steady state aerodynamic pitch torque shows that the sweep of the blades are almost balanced about zero; however, a higher sweep leads to a larger variation of the pitch torque with wind speed. The aerodynamic power of the swept blades are

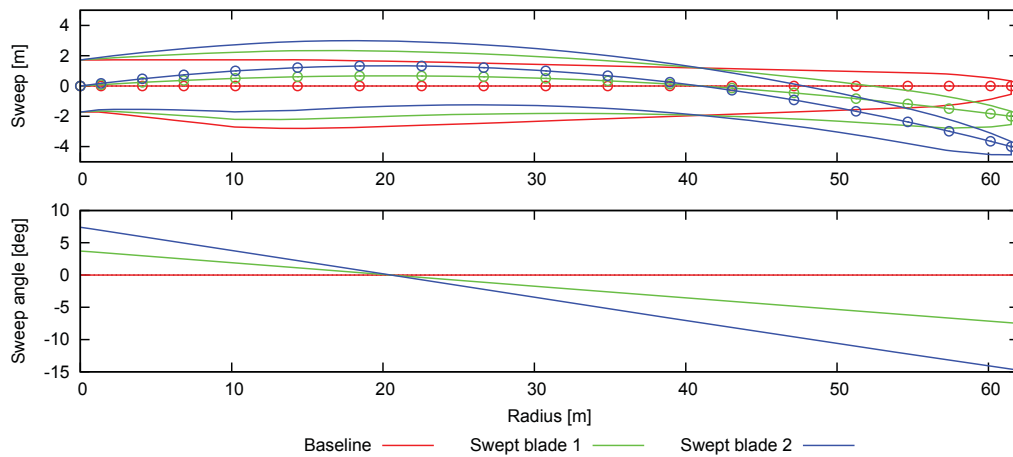


Figure 1. Baseline blade and backward blade sweep designs used in the preliminary study.

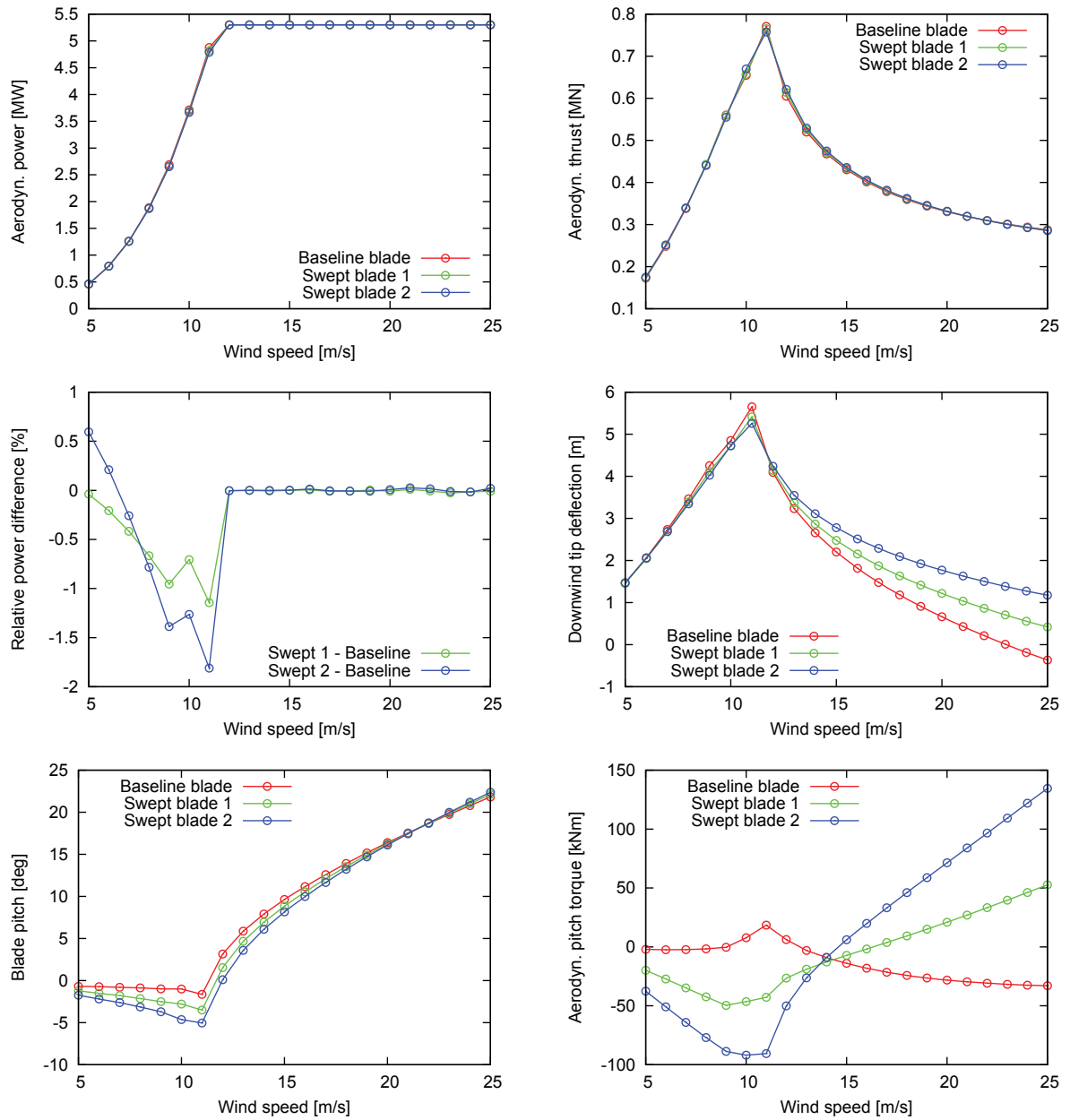


Figure 2. Mean steady state aerodynamic power, thrust, relative power difference, downwind tip deflection, pitch, and pitch torque of the three different blade designs (on a three-bladed rotor) computed for presumed optimal tip speed ratio of 8.7 up to 12.1 rpm.

slightly lower in the optimal power tracking region, which may be caused by selecting a non-optimal tip speed ratio. The maximum tip deflection and thrust at 11 m/s are slightly lower for the swept blades compared to the baseline blade. In the pitching region, the tip deflections of the three blade designs are significantly different; the swept blades remain deflected downwind, whereas the straight baseline blade bends forward.

Figure 3 shows the nonlinear deflection of the three blade designs for the steady state at 12 m/s, where all blades are producing rated power. The bending deflections are almost unchanged by the backward sweep, however, the introduced bending-torsion coupling leads to large rotations of the blade chords towards lower angles of attack, mainly on the outer half of the blade where the aerodynamic forces are largest. Note that for the swept blade 2, the static 4 m flapwise tip deflection is associated with about 5 degrees twist at the outer part of the blade, which explains the about 4 degrees difference in pitch angle.

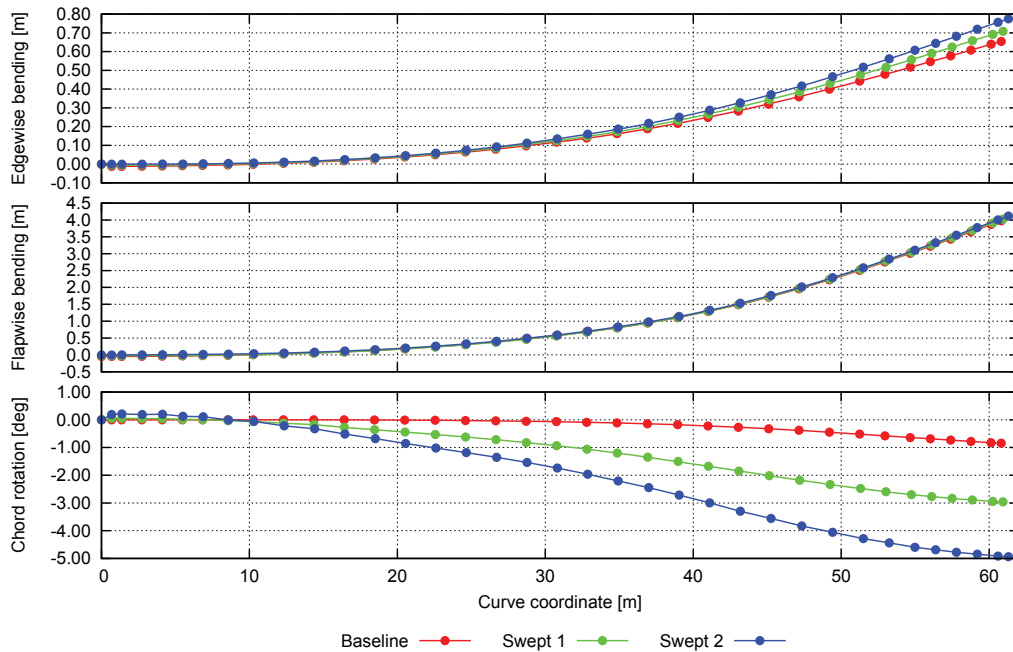


Figure 3. Mean steady state bending deflections and elastic rotations of the chord relative to the rotor plane (torsion plus geometrical effects) of the three different blade designs for operation at 12 m/s. Only plotted at the aerodynamic calculation points, thus the tip is not included in this plot.

B. Structural and aeroelastic blade modes

Figure 4 shows the structural (denoted by $\circ$) and aeroelastic (denoted by $\bullet$) modal frequencies and damping of the first flapwise and edgewise bending modes as function of wind speeds for all three blade designs. These structural and aeroelastic modal properties are obtained from eigenvalue analyses for the blades in their steady state equilibria at the different wind speeds without and with the coupling to the aerodynamic forces, respectively. The backward sweeps affect the flapwise bending mode more than the edgewise mode, for which the frequencies and damping are almost unchanged by the sweep. The aeroelastic damping of the flapwise mode are significantly reduced, whereas the aeroelastic frequencies are increased by the sweep. The aeroelastic frequency of the baseline blade is close to its corresponding structural frequency of the flapwise mode, which shows that the aerodynamic forces add stiffness to the swept blades. To explain these observations, it is necessary to consider the bending-torsion couplings in the mode shapes. The large changes in damping at rated wind speed is caused by operation close to stall on parts of the blade.

Figure 5 shows the absolute magnitudes of bending deflection and local torsion (negative towards feathering) with its phase in the first structural (denoted by $\circ$) and aeroelastic (denoted by $\bullet$) flapwise and edgewise blade bending modes at 12 m/s, where the mode shapes are normalized to have a maximum tip deflection of 1 m. First, it is noted that the phase of the torsional component in the structural modes are the anti-phase with the dominating bending component of both bending modes, except for the straight baseline blade where the phase is around 120° due to a very small torsional component. Hence, the swept blades are undergoing torsion towards feathering for downwind deflections in their first flapwise modes, whereas all three blades undergo torsion towards feathering for bending towards the leading edgewise in their first edgewise modes due to the large steady state flapwise deflection.

For the aeroelastic edgewise mode, the magnitude and phase of the torsional component are only slightly changed due to the aeroelastic coupling with the aerodynamic forces. For the aeroelastic flapwise mode, however, this aeroelastic coupling creates a significant torsional component of the straight baseline blade which is closer to 90° out of phase with the flapwise deflection. This torsional component and its phase can be explained by the direct link of the flapwise velocity over the angle of attack to the aerodynamic moment about the elastic axis and coinciding shear center along the blade. This direct link is also the reason for the high aerodynamic damping of the flapwise bending mode of the pitch-regulated blade.

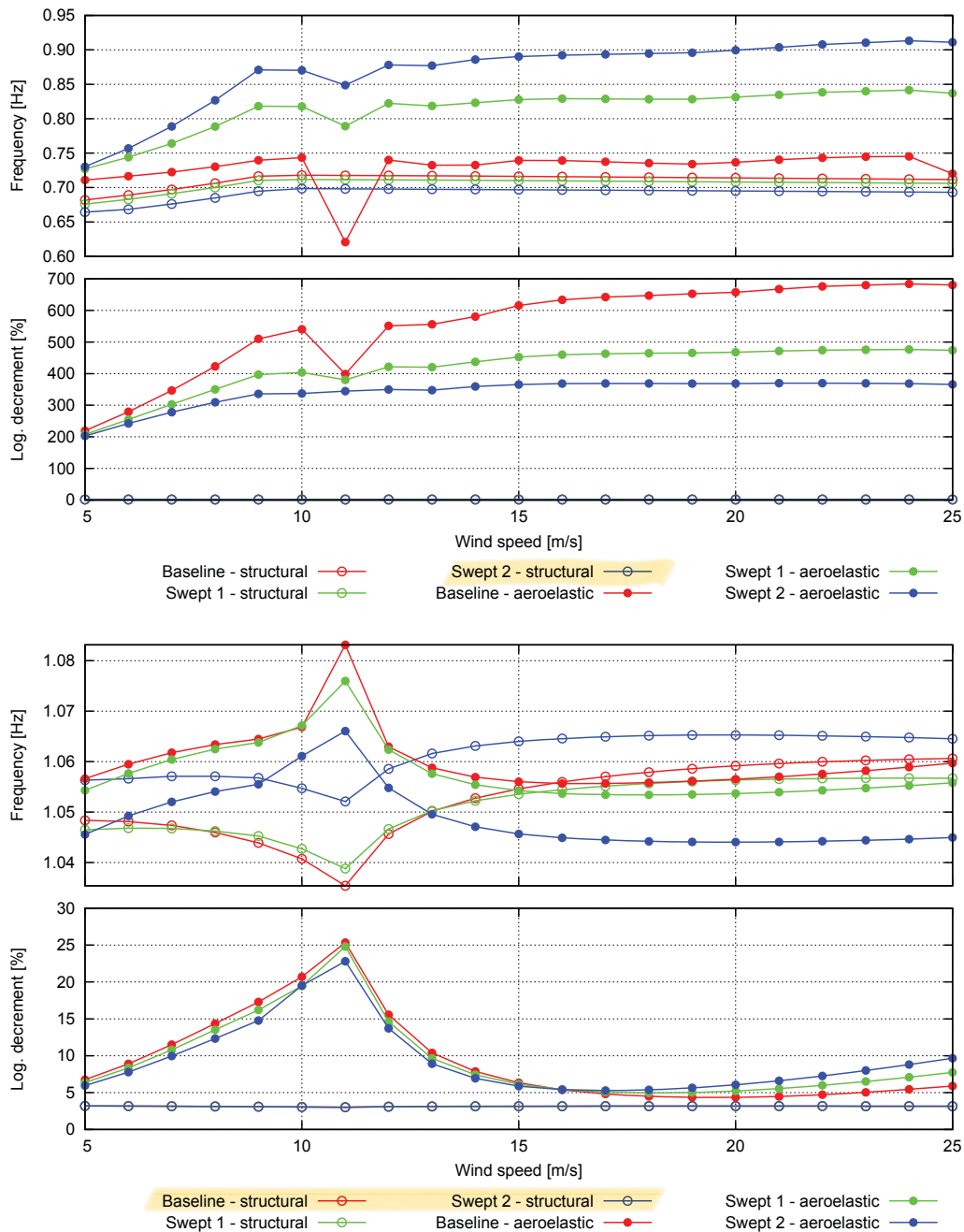


Figure 4. Structural (denoted by $\circ$) and aeroelastic (denoted by $\bullet$) modal frequencies and damping in logarithmic decrements of the first flapwise (top) and first edgewise (bottom) mode shapes for the three blade designs as function of wind speeds.

The aeroelastic coupling of the blade motion with the aerodynamic forces also changes the torsional components of the swept blades by increasing their amplitudes and phases from 180° towards 90° in the tip region; however, the phases remain close 180° over a large part of the blades, which can be explained by the structural coupling of downwind flapwise deflection and torsion towards feathering as seen in their corresponding structural modes. The decreased aeroelastic damping of the flapwise modes of the swept blades observed in Figure 4 can now be explained the decreased torsional component which is 90° out of phase with the flapwise motion and therefore adds to the damping. Opposite, the increased aeroelastic flapwise frequencies of the swept blades can be explained by the structural coupling that creates torsion towards feathering for downwind flapwise deflection, whereby the aerodynamic lift (pointing mainly downwind) is reduced when the blade deflecting downwind, and therefore acts as a stiffening restoring force.

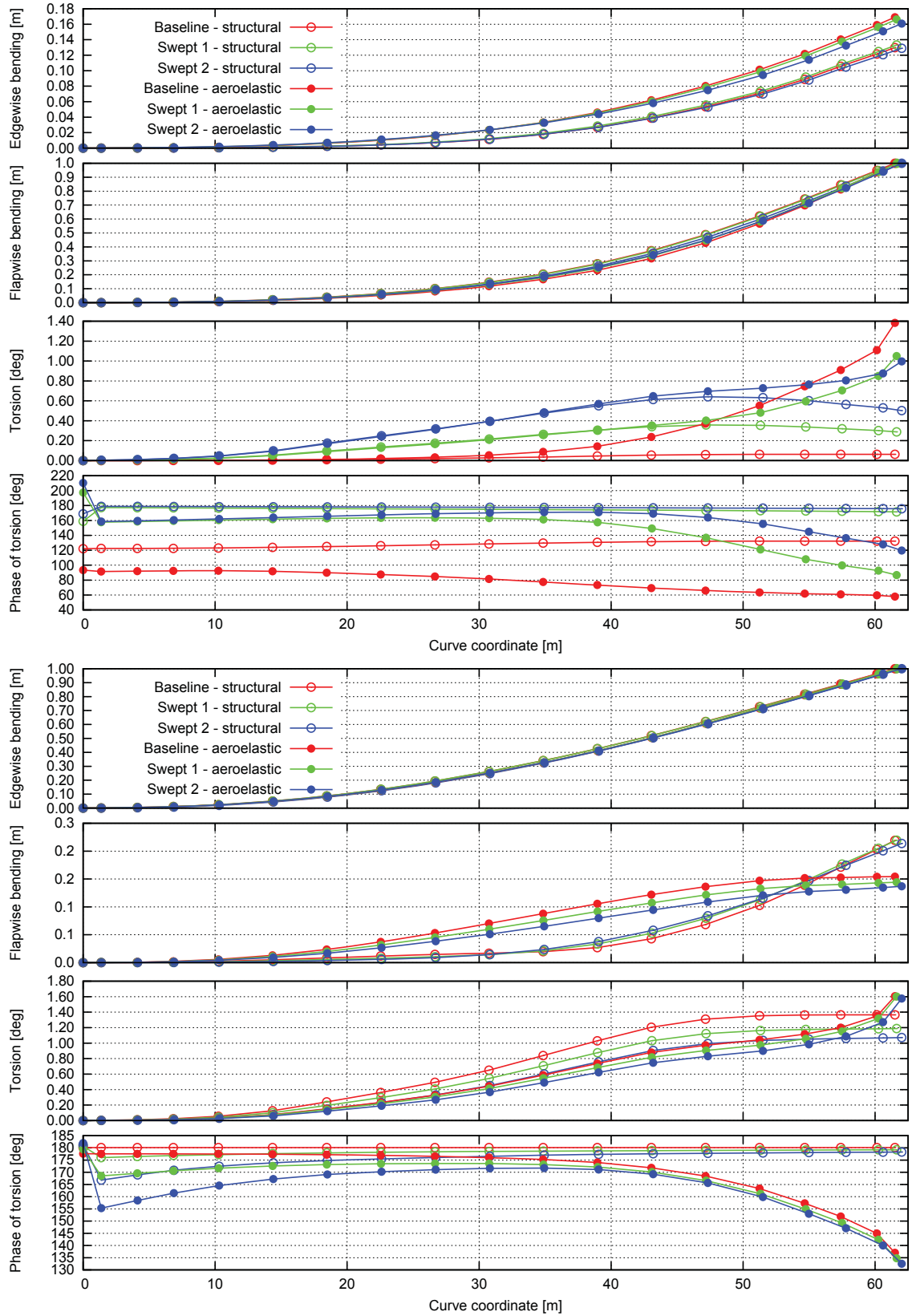


Figure 5. Absolute magnitudes of bending and torsion about the local blade tangent (negative towards feathering) in the first flapwise (top) and first edgewise (bottom) structural (denoted by $\circ$) and aeroelastic (denoted by $\bullet$) mode shapes for the three blade designs under steady state operation at 12 m/s. The phase of the torsional component is relative to the maximum tip deflection motion which magnitude is normalized to 1 m. Note that sign changes of the magnitudes occur whenever the phase of the particular component crosses $\pm 90^\circ$.

C. Transfer functions from mean wind to flapwise moment

Figures 6 and 7 show Bode plots of the transfer function from harmonic mean speed variation to flapwise and edgewise blade root bending moments, respectively, for steady state operation at mean wind speeds 5, 10, 15, and 20 m/s. The frequency range is 0 to 2 Hz in which the first flapwise and edgewise bending modes have their modal frequencies according to Figure 4. The edgewise mode is clearly excited due to its relative low total damping, whereas the resonance peak for the flapwise mode is unclear due to its high aerodynamic damping. It is noted that the frequency response of the flapwise blade root bending moments in the frequency range below the first flapwise modes is reduced by the backward sweep. Although, the real spanwise dependent excitation of the blades by the inflow turbulence is not accurately modeled by a variation of the mean wind, the reduced frequency responses can qualitatively explain the reductions of flapwise blade root bending moments observed in all previous studies of backward swept blades, because the main energy of the excitation by rotational sampled turbulence will be in the 1P to 3P frequency range¹³ also plotted in Figures 6 and 7. Similarly, the slight increase of edgewise blade root bending moments found by Verelst and Larsen⁴ can be qualitatively explained by the increased responses due to the sweep in the same frequency range.

The cause of the reduced frequency response of the flapwise blade root bending moments below the frequency of the first flapwise mode is that this frequency is increased by the backward sweep as explained in the previous section. Opposite, the slightly increased frequency response of the edgewise blade root bending moments is caused by a slight decrease in aeroelastic damping of the first edgewise bending mode (cf. Figure 4).

D. Flutter speed in overspeed situations

To investigate the effect of the sweep on the risk of flutter in an overspeed situation, the blades are operated at a constant pitch angle of 0° and a constant tip speed ratio of $\sqrt{99}$, whereby the relative inflow speed at the tip becomes ten times the wind speed. For each wind speed, the steady state deflection of the blade is computed and the aeroelastic eigenvalue problem for small blade vibrations about it is solved. Figure 8 shows the aeroelastic frequencies and

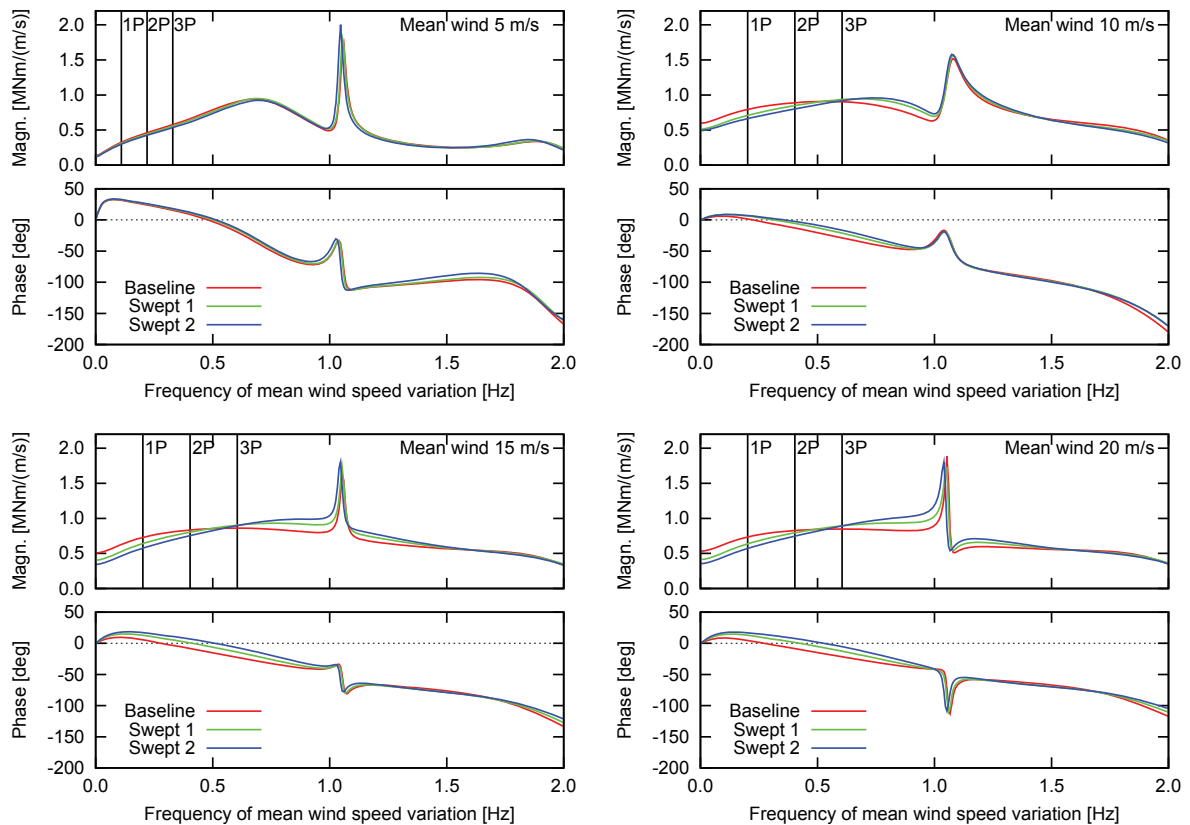


Figure 6. Bode plots of the transfer function from spanwise mean speed variation to flapwise blade root bending moment for steady state operation at mean wind speeds 5, 10, 15, and 20 m/s.

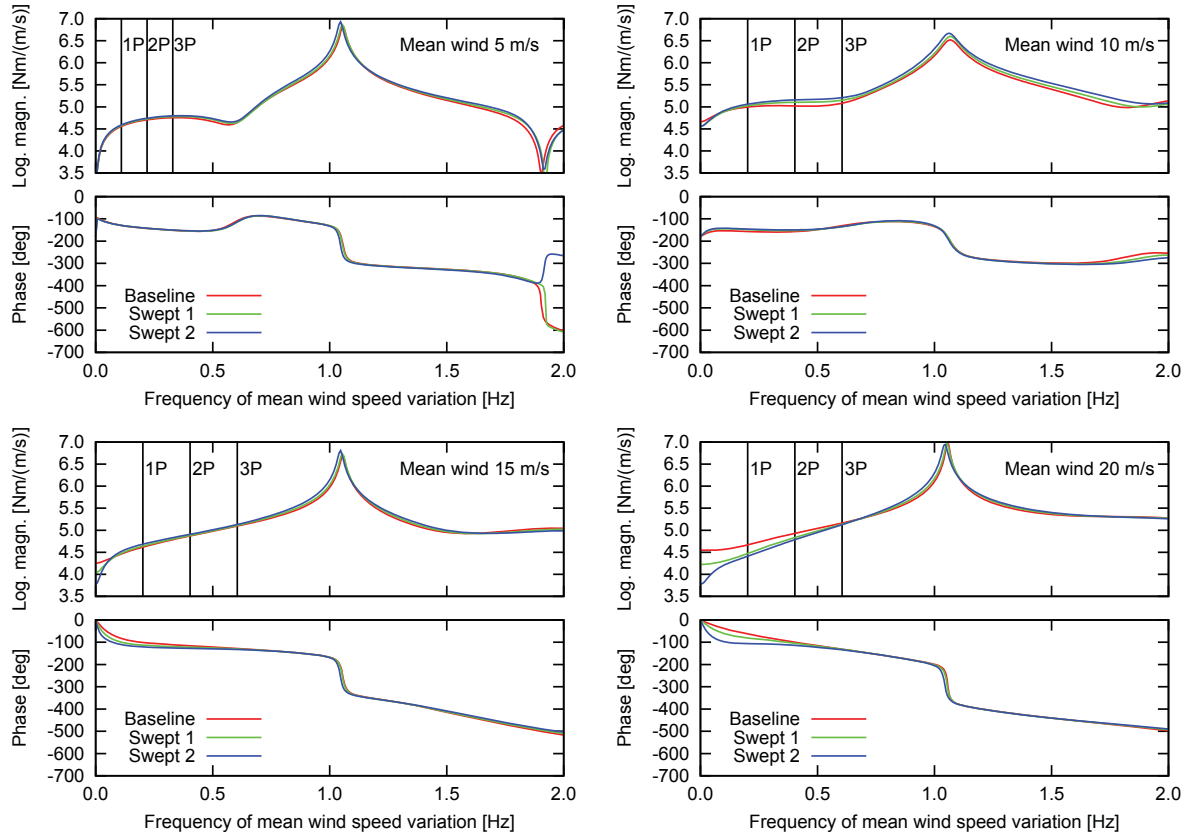


Figure 7. Bode plots of the transfer function from spanwise mean speed variation to edgewise blade root bending moment for steady state operation at mean wind speeds 5, 10, 15, and 20 m/s.

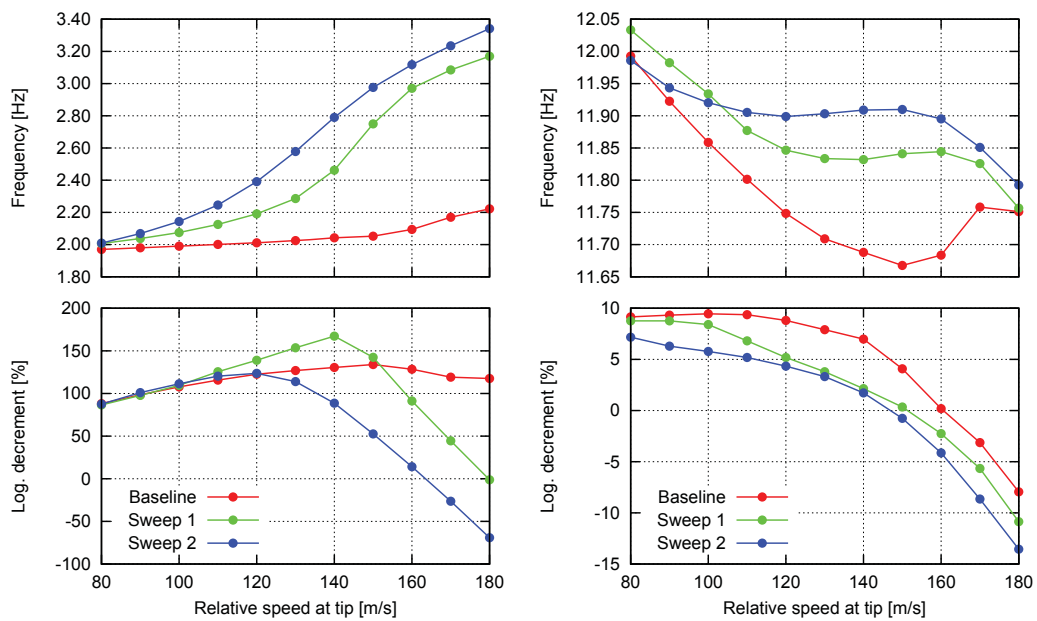


Figure 8. Aeroelastic modal frequencies (top) and damping in logarithmic decrements (bottom) of the second flapwise (left) and second torsional (right) mode shapes for the three blade designs as function of wind speeds.

damping of the lowest damped modes of the three blade designs for wind speeds in the range from 8 to 18 m/s, corresponding to relative inflow speeds at the tip from 80 to 180 m/s. The first mode to become negatively damped is the second torsional mode with a frequency close to 12 Hz. This instability occurs at around 145 m/s for the swept blade 2 and 175 m/s for the baseline blade. Animations of the corresponding mode shape (not shown) indicate flutter-like motions of some airfoils along the blade spans; but not all sections seem to contribute to instability as also seen by the moderate levels of negative damping. Animations of the second flapwise modes that become negatively damped for the swept blade at around 175 m/s and 180 m/s, show motions that much more like classical flutter, which also can be seen by the characteristic large levels of negative damping. The baseline blade also has this negative damped flutter mode but at a much higher flutter speed.

V. Conclusion

The effects of the nonlinear geometric coupling between torsion and flapwise bending of backward swept blades have been investigated using a nonlinear co-rotational finite beam element formulation coupled with the Blade Element Momentum method including an unsteady aerodynamic model. A linearization of this aeroelastic model about the steady state blade deflection has enabled the computation of frequencies, damping and mode shapes of the aeroelastic blade modes from eigenvalue analysis, and the computation of transfer functions from mean wind speed variations to variations in the blade root bending moments. Using the blade of the 5 MW NREL reference turbine as a baseline, it has been shown that the backward sweep creates torsion towards feathering for downwind flapwise deflection in the first flapwise bending mode. This torsional component was shown to cause the frequency of the first aeroelastic flapwise bending mode to increase compared to the corresponding structural frequency, and this increase becomes larger for larger sweep. The frequency response of the flapwise blade root moment from wind excitation was shown to decrease below the increased first flapwise frequency which explains the reduced flapwise loads found in previous studies of backward swept blades. Computations of the classical flutter speed showed that it decreases with backward sweep.

Acknowledgments

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